SUPERSTORE SALES ANALYSIS REPORT

Data Science Portfolio Project

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Tools Used: Python | Jupyter Notebook | Pandas | Matplotlib | Seaborn

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Dataset: Superstore Sales Dataset

Records Analyzed: 9,994 rows × 21 columns

Period Covered: 2014 – 2017

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2. EXECUTIVE SUMMARY

This report presents a comprehensive analysis of the Superstore Sales dataset covering the period **2014 to 2017.** The dataset contains **9,994** transactions across **793** customers and **5,009** unique orders, generating a total revenue of **$2,297,200.86** and a total profit of **$286,397.02**, reflecting an overall profit margin of **12.03%.**

The analysis reveals that Technology is the highest performing category in both sales and profit, while the West region leads all geographic areas in revenue contribution. The top customer by total purchases is Sean Miller.

However, the analysis also uncovers critical business concerns. Three sub-categories Tables, Bookcases and Supplies are operating at a loss. Furthermore, a strong negative relationship was found between discount levels and profit, indicating that the company's current discounting strategy is significantly hurting its bottom line.

Sales patterns show clear seasonality, with peak performance in August and November driven by back-to-school and holiday shopping, while January consistently records the lowest sales across all years.

Based on these findings, this report provides actionable recommendations to help the business improve profitability, optimize its discount strategy and focus resources on its strongest markets and products.

3. DATA OVERVIEW

3.1 Dataset Description

The dataset used in this analysis is the Superstore Sales Dataset, a widely recognized dataset in the data science community representing transactional sales data from a fictional US-based retail superstore. The data spans four years from **2014 to 2017** and covers three main product categories sold across four geographic regions of the United States.

3.2 Dataset Structure

The dataset contains **9,994 rows** and **21 original columns**, which were expanded to **26 columns** after feature engineering. The table below summarizes the key variables used in this analysis:

Column	Type	Description
Order.Date	Date	Date the order was placed
Ship.Date	Date	Date the order was shipped
Ship.Mode	Text	Shipping method chosen
Customer.Name	Text	Name of the customer
Segment	Text	Customer segment — Consumer, Corporate, Home Office
Region	Text	Geographic region — East, West, North, South
Category	Text	Product category — Furniture, Office Supplies, Technology
Sub.Category	Text	Product sub-category
Sales	Number	Revenue generated per transaction
Profit	Number	Profit earned per transaction
Discount	Number	Discount applied per transaction
Quantity	Number	Number of items ordered

3.3 Data Preparation

Before analysis, the following preparation steps were performed:

- Date columns **Order.Date** and **Ship.Date** were converted from text format to proper date format
- The dataset was checked for missing values — **no missing values were found**
- The dataset was checked for duplicate records — **no duplicates were found**
- Five new columns were engineered from existing data to support deeper analysis:

New Column	Description
Order.Month	Month extracted from Order.Date
Order.Year	Year extracted from Order.Date
Order.Quarter	Quarter extracted from Order.Date
Ship.Delay	Number of days between order and shipment
Profit.Margin	Profit as a percentage of Sales

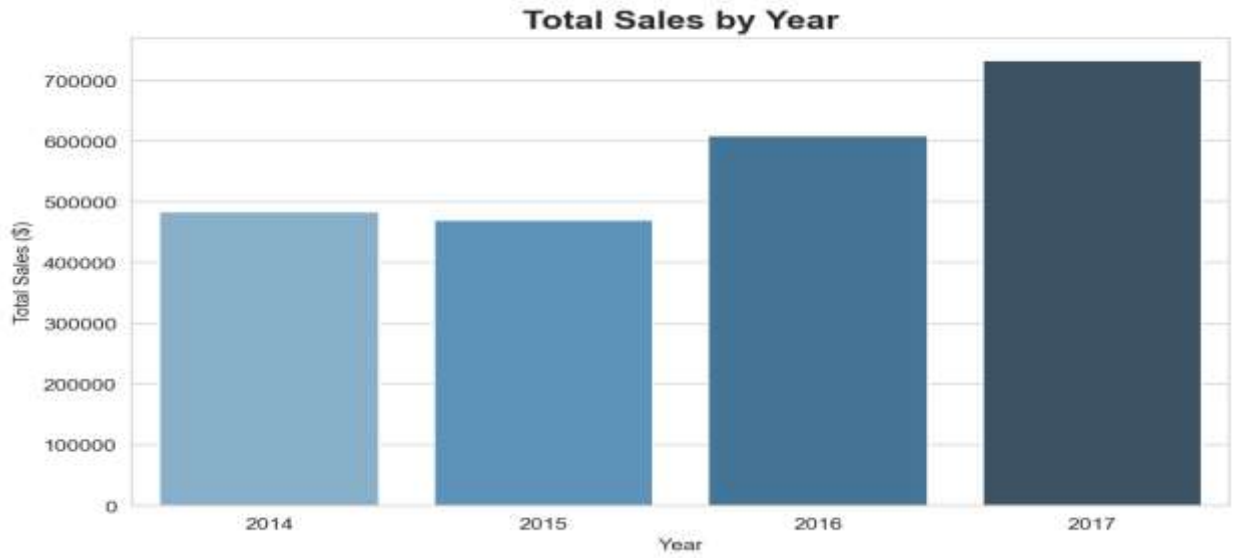
3.4 Key Dataset Statistics

Metric	Value
Total Sales Revenue	$2,297,200.86
Total Profit	$286,397.02
Overall Profit Margin	12.03%
Total Unique Orders	5,009
Total Unique Customers	793
Maximum Shipping Delay	7 days
Minimum Shipping Delay	0 days

4. KEY FINDINGS

4.1 Sales Performance Over Time

4.1.1 Annual Sales Trend



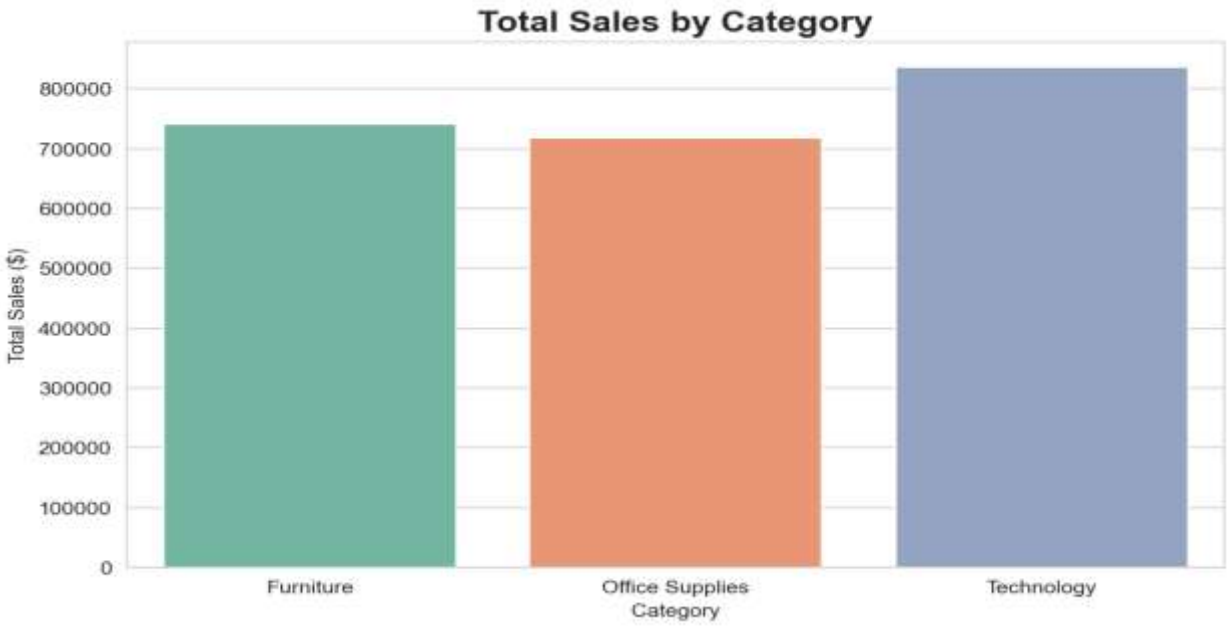
The analysis of annual sales reveals a consistent upward trend throughout the period. The year 2017 recorded the highest total sales, demonstrating strong business growth year over year. The year 2015 recorded the lowest sales, which is attributable to incomplete data coverage for that period. This consistent growth pattern suggests a healthy and expanding customer base.

4.1.2 Monthly Sales Trend

Sales data reveals clear seasonal patterns across all years. January consistently records the lowest sales figures, reflecting a post-holiday spending slowdown among consumers. Significant peaks are observed in August and November, driven by back-to-school purchasing and the onset of the holiday shopping season respectively. These seasonal patterns remain consistent across all four years, confirming their reliability for future business planning.

4.2 Category & Product Performance

4.2.1 Sales by Category

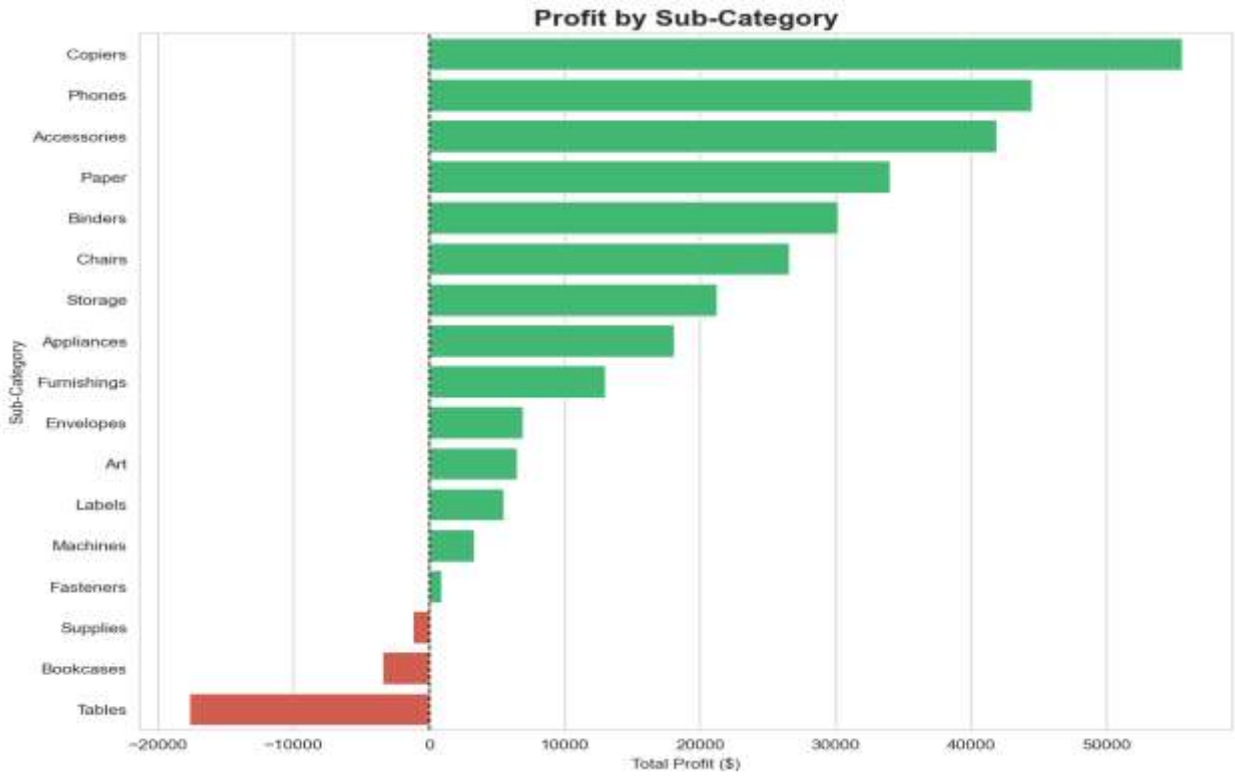


Technology emerges as the leading category in total sales, followed by Furniture and Office Supplies respectively. Despite Office Supplies recording the lowest total sales value, it is important to note that this category generates a high volume of small transactions, suggesting strong repeat purchasing behavior among customers.

4.2.2 Profit by Category

The profit analysis confirms Technology as the strongest performing category in both sales and profitability. This dual leadership in sales and profit makes Technology the most strategically valuable category for the business. Furniture, despite being the second highest in sales, shows relatively weaker profit performance a pattern that warrants further investigation at the sub-category level.

4.2.3 Profit by Sub-Category



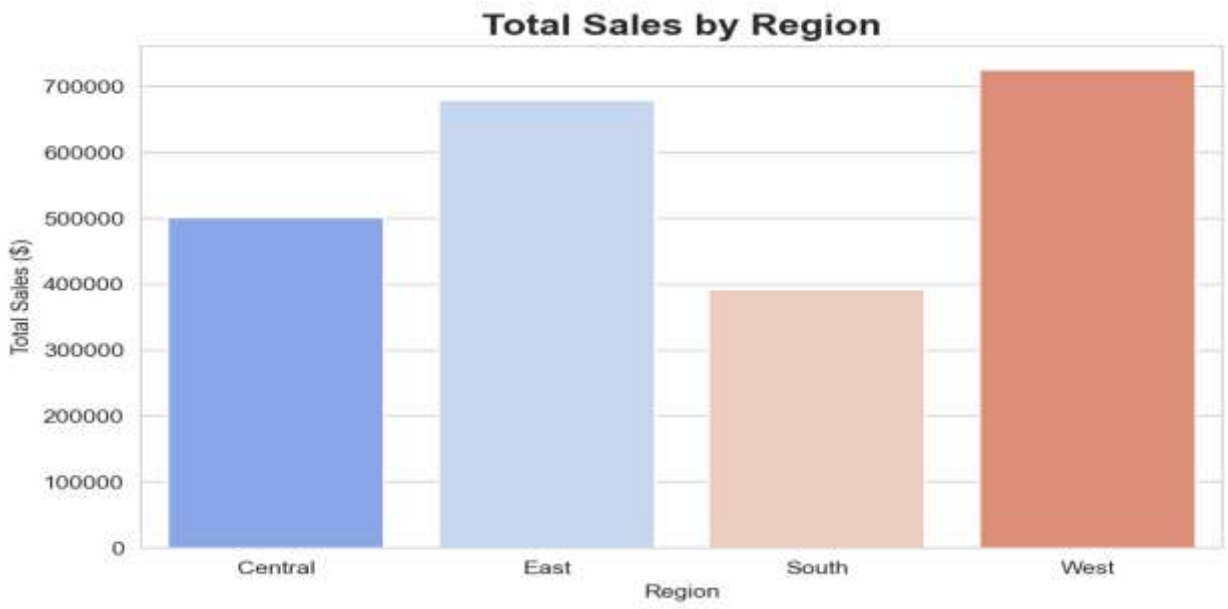
Drilling down to the sub-category level reveals critical insights. Copiers record the highest profit of all sub-categories, confirming their status as the single most valuable product line in the entire business. However, three sub-categories are operating at a financial loss:

Sub-Category	Loss Severity	Recommended Action
Tables	High	Urgent pricing review required
Bookcases	Moderate	Discount strategy review needed
Supplies	Low	Monitor and reassess quarterly

These loss-making sub-categories represent a significant threat to overall business profitability and require immediate management attention.

4.3 Geographic Performance

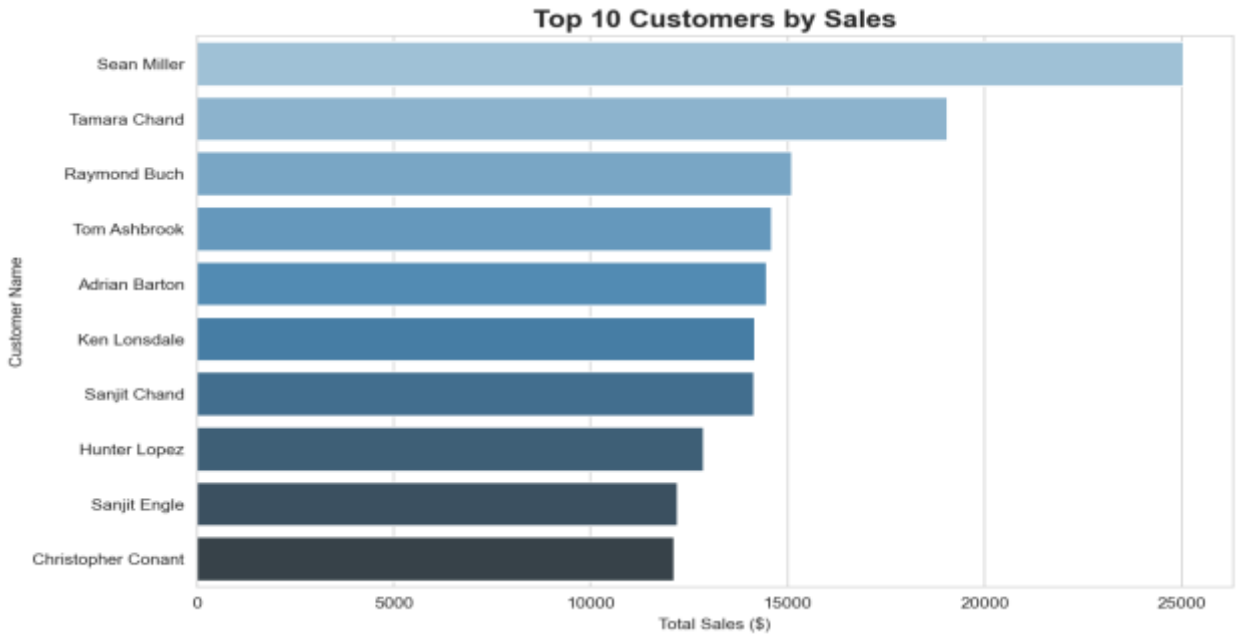
4.3.1 Sales by Region



The West region leads all geographic areas with the highest total sales, establishing it as the company's most important market. The South region records the lowest sales performance among all four regions. This geographic disparity suggests an opportunity to investigate what drives West region success and replicate those strategies in underperforming regions, particularly the South.

4.4 Customer Analysis

4.4.1 Top 10 Customers by Sales



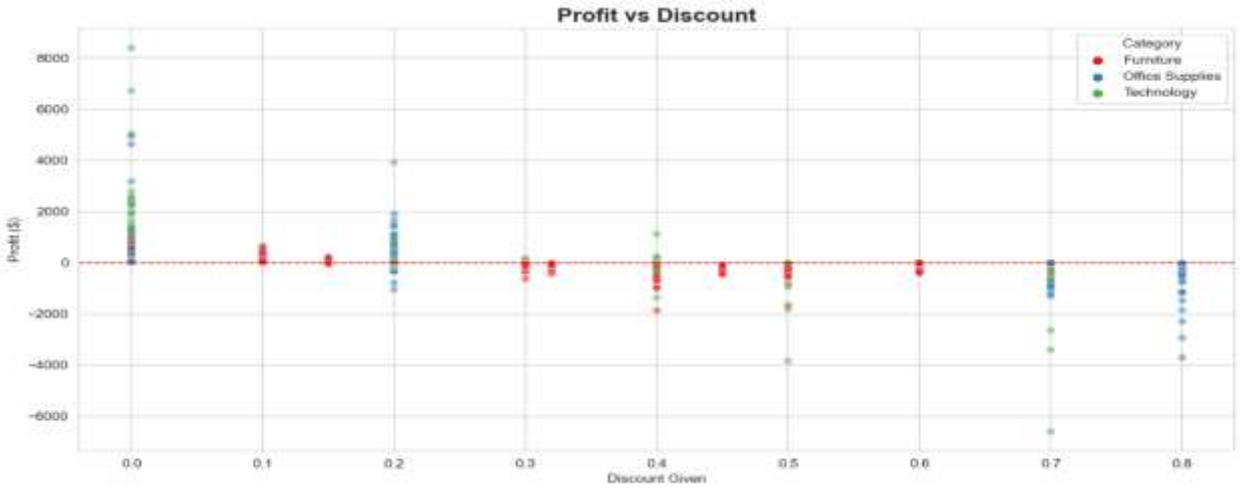
The top 10 customers represent a disproportionately high share of total revenue, with Sean Miller ranking as the number one customer by total purchases. This concentration of revenue among a small group of customers highlights the importance of implementing a customer retention strategy to protect these high-value relationships. Losing even one top customer could have a measurable impact on annual revenue

4.4.2 Profit Margin by Customer Segment

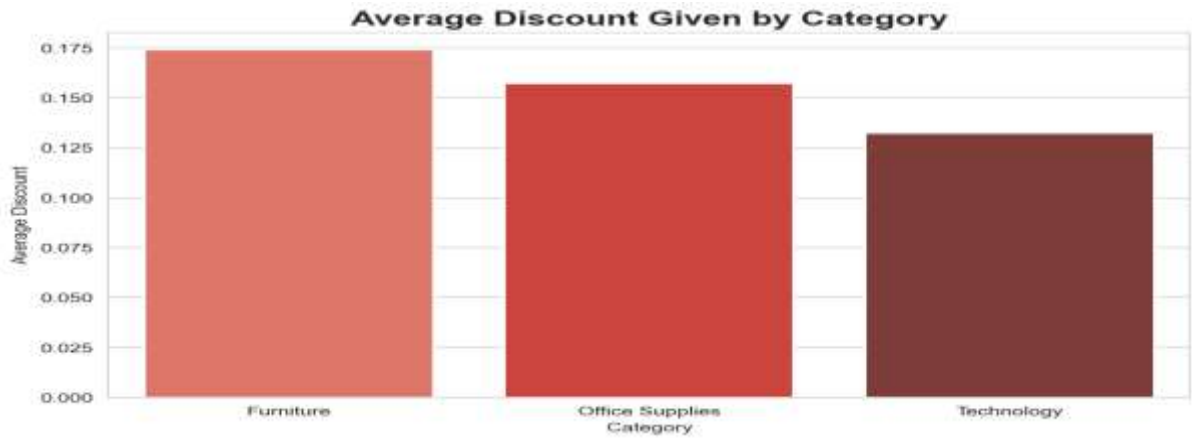
Analysis of the three customer segments reveals differences in profitability across Consumer, Corporate and Home Office groups. Understanding which segment delivers the highest profit margin allows the business to prioritize its sales and marketing efforts toward the most financially rewarding customer groups.

4.5 Discount Impact Analysis

4.5.1 Profit vs Discount



4.5.2 Average Profit at Each Discount Level



The discount analysis reveals one of the most critical findings in this report. When no discount is applied, the majority of transactions generate positive profit, particularly in the Technology category. However as discount levels increase, profit rapidly deteriorates and crosses into negative territory. This demonstrates a strong negative relationship between discounting and profitability, meaning the company's current discounting strategy is actively destroying value rather than driving sustainable growth.

5. BUSINESS RECOMMENDATIONS

Based on the findings presented in this report, the following strategic recommendations are proposed to improve business performance and profitability:

5.1 Revise the Discounting Strategy

Finding: The analysis clearly demonstrates that high discount levels directly cause profit losses across all categories, with the effect most severe in the Furniture category.

Recommendation:

- Immediately cap all discounts at a maximum of **20%** across all product categories
- Eliminate discounts entirely on already loss-making sub-categories Tables, Bookcases and Supplies
- Introduce a **discount approval process** where any discount above 15% requires management sign-off
- Replace blanket discounting with targeted promotions tied to specific customer segments or seasonal events

***Expected Impact*:** Reducing excessive discounting is the single fastest way to improve profit margins without changing sales volume.

5.2 Invest More in Technology Category

Finding: Technology leads the business in both sales and profit, with Copiers being the single most profitable sub-category in the entire product range.

Recommendation:

- Increase marketing budget allocation toward Technology products
- Expand the Copiers product range with newer models and accessories
- Train sales teams to upsell Technology products during customer interactions
- Consider offering Technology bundles to increase average order value

Expected Impact: Doubling down on the strongest category will accelerate revenue and profit growth simultaneously.

5.3 Address Loss-Making Sub-Categories

***Finding*:** Tables, Bookcases and Supplies are consistently generating losses, dragging down overall profitability despite contributing to sales volume.

Recommendation:

Sub-Category	Recommended Action	Priority
Tables	Re-evaluate supplier costs and increase base price	High
Bookcases	Reduce discount levels immediately	High
Supplies	Monitor for two quarters then reassess	Medium

- Conduct a detailed cost analysis on Tables to determine if the product line is viable
- If Tables cannot be made profitable within two quarters, consider **discontinuing the product line**
- Negotiate better supplier pricing for Furniture sub-categories to improve margins

Expected Impact: Eliminating losses in these three sub-categories could recover a significant portion of lost profit annually.

5.4 Capitalize on Seasonal Patterns

Finding: Sales peak in August and November while January is consistently the slowest month across all years.

Recommendation:

- Plan major **stock increases** in July and October to prepare for August and November peaks

- Launch a dedicated **January promotion campaign** targeting existing customers to combat the seasonal slowdown
- Introduce a loyalty rewards program that encourages repeat purchases during slow months
- Use the November peak to clear slow-moving Furniture inventory with time-limited offers

Expected Impact: Smoothing seasonal fluctuations will create more consistent cash flow throughout the year.

5.5 Focus on West Region and Fix South Region

***Finding*:** The West region consistently outperforms all other regions while the South region records the lowest sales performance.

Recommendation:

- Conduct a deep dive analysis into what drives West region success identify top sales channels, customer demographics and product preferences
- Replicate the West region strategy in the South region through targeted marketing campaigns
- Assign dedicated sales representatives to the South region to build customer relationships
- Offer South region customers introductory promotions to stimulate demand

Expected Impact: Bringing the South region performance closer to the West region level could add significant revenue without acquiring new customers.

5.6 Protect and Grow Top Customer Relationships

Finding: A small group of customers including Sean Miller as the top customer generates a disproportionately high share of total revenue.

Recommendation:

- Implement a **VIP customer program** for the top 20 customers by revenue
- Assign dedicated account managers to the top 10 customers
- Offer exclusive early access to new products and special pricing for VIP customers

- Conduct quarterly check-in calls with top customers to understand their evolving needs

Expected Impact: Retaining the top 10 customers alone protects a significant portion of annual revenue and reduces customer churn risk.

6. CONCLUSION

This report has presented a comprehensive data-driven analysis of the Superstore Sales dataset covering four years of transactional data from 2014 to 2017. Through systematic exploration of sales performance, product profitability, customer behavior, geographic distribution and discount impact, several critical insights have been uncovered that carry significant implications for business strategy.

The analysis confirms that the business is fundamentally healthy, generating **$2,297,200.86 in total revenue** and **$286,397.02 in total profit** across **5,009 orders** from **793 unique customers**. The consistent year-over-year sales growth demonstrates a strong and expanding customer base with reliable seasonal demand patterns.

However the analysis also reveals clear areas of concern that require immediate attention. The negative impact of excessive discounting on profitability is the single most urgent issue identified, as it is actively eroding margins across all product categories. The loss-making performance of Tables, Bookcases and Supplies sub-categories further compounds this challenge and demands a strategic pricing and cost review.

On the opportunity side, Technology particularly Copiers represents a powerful growth engine that the business should prioritize in its sales and marketing investments. The West region's consistent outperformance provides a proven blueprint that can be adapted and replicated in underperforming regions, especially the South.

By implementing the six recommendations outlined in this report revising the discount strategy, investing in Technology, addressing loss-making sub-categories, capitalizing on seasonal patterns, developing the South region and protecting top customer relationships the business is well positioned to significantly improve its profitability and sustain its growth trajectory in the years ahead.